

Tech & Teach

Digital Chip Design Program

RTL Design

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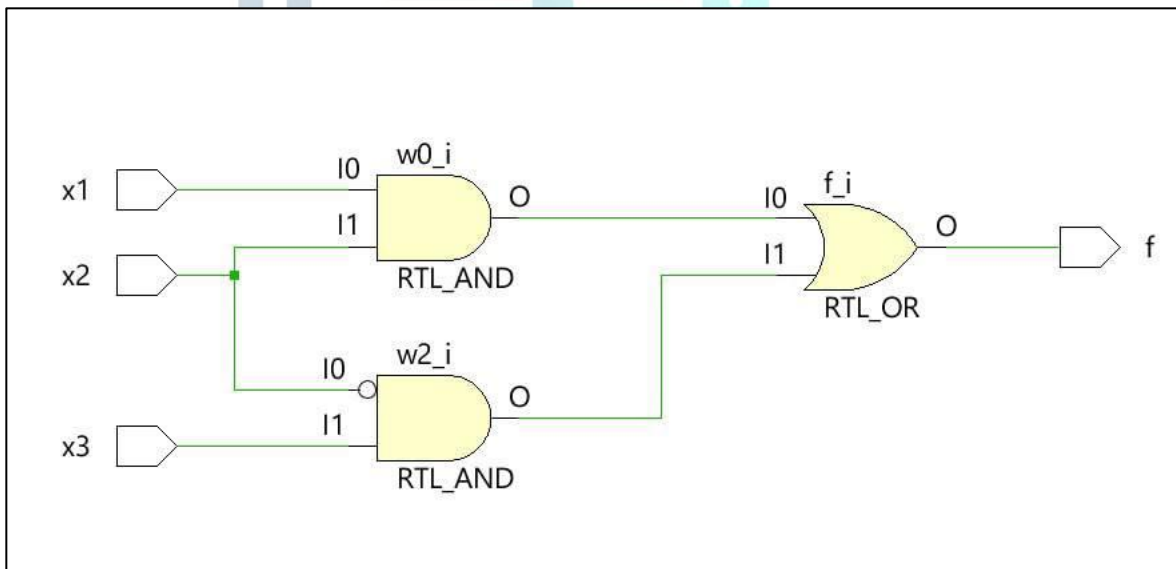
Task #1

29/6/2026

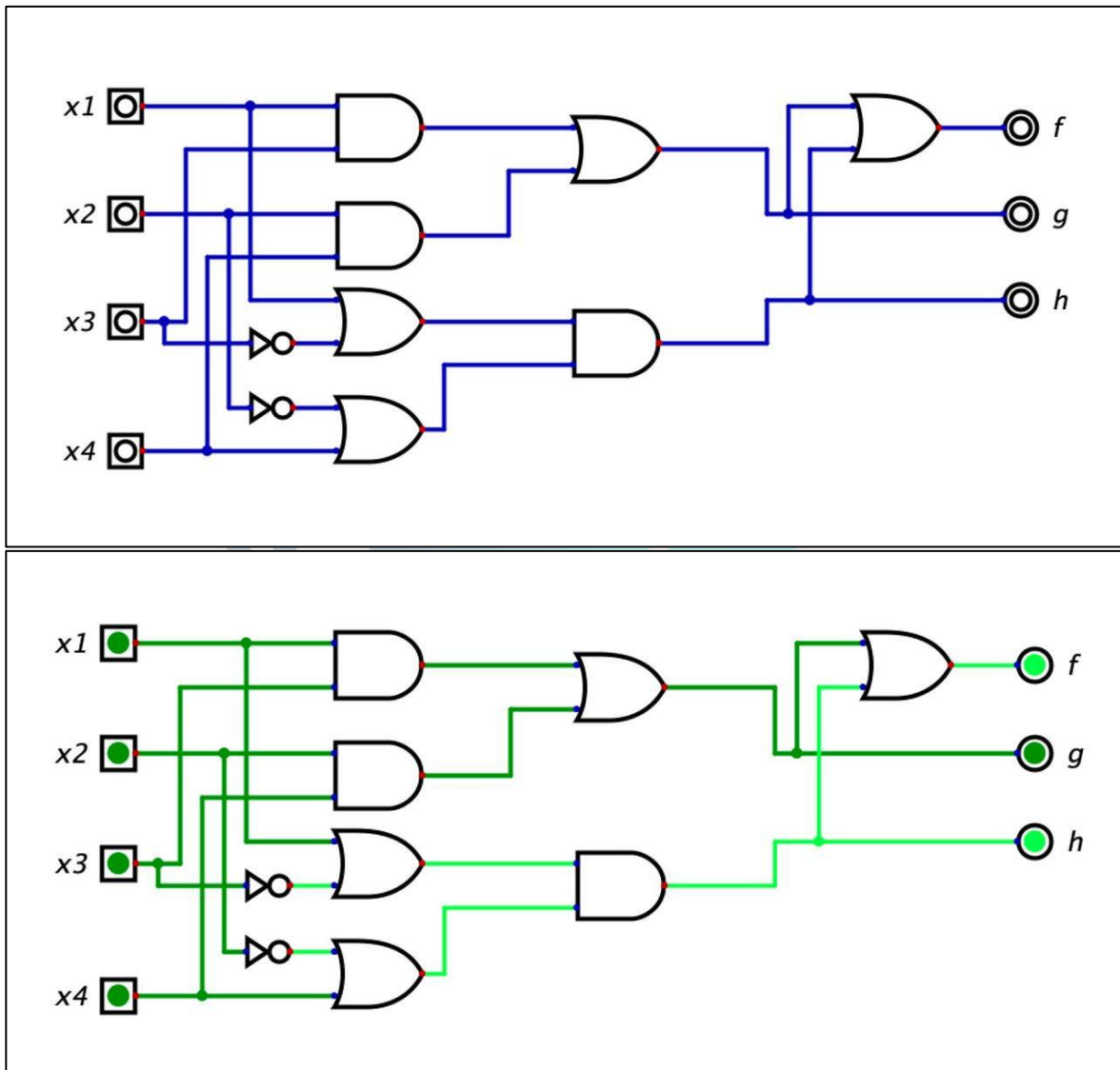
Gate-Level

Task #1.1

```
module task1_1(  
  input logic x1,  
  input logic x2,  
  input logic x3,  
  output logic f  
);  
  logic w0, w1, w2;  
  and(w0, x1, x2);  
  not(w1, x2);  
  and(w2, w1, x3);  
  or(f, w0, w2);  
endmodule
```



Task #1.2

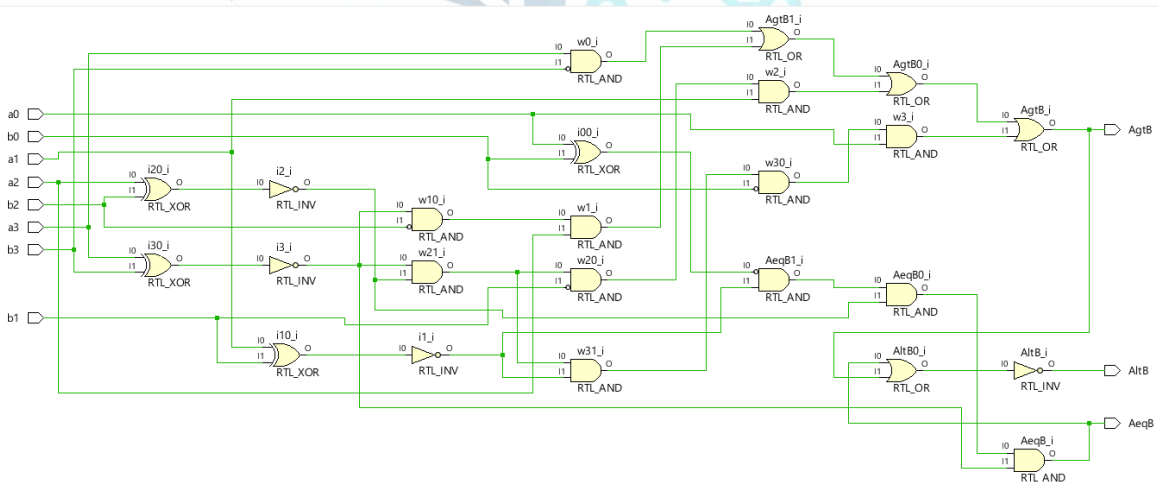


Task #1.3

```

1  module task11(
2      input logic a3,
3      input logic b3,
4      input logic a2,
5      input logic b2,
6      input logic a1,
7      input logic b1,
8      input logic a0,
9      input logic b0 ,
10     output logic AeqB,AltB,AgtB
11 );
12     logic nb0, nb1, nb2, nb3;
13     logic i0, i1, i2, i3,L0,L1;
14     logic w0,w1,w2,w3;
15     xnor (i3,a3,b3);
16     xnor (i2,a2,b2);
17     xnor (i1,a1,b1);
18     xnor (i0,a0,b0);
19     not (nb3,b3);
20     and(w0,a3,nb3);
21     not(nb2,b2);
22     and(w1,i3,nb2,a2);
23     not(nb1,b1);
24     and(w2,i3,i2,nb1,a1);
25     not(nb0,b0);
26     and(w3,i3,i2,i1,nb0,a0);
27     and (AeqB,i0,i1,i2,i3);
28     or (AgtB,w0,w1,w2,w3);
29     nor (AltB,AeqB,AgtB);
30
31 endmodule

```



Work distribution

Section	ROAA	RAND	LANA	LARA	RAWAN	RENAD
Report	✓	✓				
Task 1.1					✓	✓
Task 1.2			✓	✓		
Task 1.3	✓	✓	✓	✓	✓	✓